COURSE TITLE	**FOUNDATIONS OF MACHINE LEARNING**
COURSE CODE	**04BA0501**
COURSE CREDITS	**3**

Objective:

1 N/A

Course Outcomes: After completion of this course, student will be able to:

1 Build an understanding of the basic concepts, types, and challenges in machine learning.
2 Apply regression and classification models for predictive analytics.
3 Evaluate and improve the model, performance using resampling and regularization techniques.
4 Develop tree-based and support vector machine models for business decision- making.
5 Apply unsupervised learning techniques like PCA and clustering in real-world scenarios.

Pre-requisite of course: Basic knowledge of statistical concepts.

Teaching and Examination Scheme

Theory Hours	Tutorial Hours	Practical Hours	ESE	IA	CSE	Viva	Term Work
3	0	0	50	30	20	0	0

Contents : Unit	Topics	Contact Hours
1	**Unit 1: Foundations of Machine Learning in Business** Introduction to Machine Learning, Learning Systems and Real-world ML Applications, Importance of ML in Business Analytics,, Variable Types and Terminology, Function Approximation, Concept of Learning and Types of Machine Learning, Curse of Dimensionality, Bias-Variance Tradeoff, Overfitting and Underfitting,, Model Selection Techniques, No Free Lunch Theorem	10
2	**Unit 2: Predictive Modeling – Regression and Classification** Linear Regression, Simple Linear Regression,, Model Accuracy: R^2, Logistic Regression, Linear Discriminant Analysis (LDA), Bayes Theorem, Naïve Bayes Classifier	9
3	**Unit 3: Resampling, Regularization, and Tree-Based Techniques** Model Evaluation and Resampling Methods,, Cross-Validation: LOOCV, k-Fold, The Bootstrap Method,, Principal Component Regression (PCR),, Decision Trees for Classification and Regression, Advantages and Disadvantages of Decision Trees,, EnsembleMethods: Bagging, Random Forest, Boosting	10

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Contents : Unit	Topics	Contact Hours
4	**Unit 4: Support Vector Machines and Applications** Concepts of Margins and Hyperplanes,, Maximum Margin Classifier, Support Vector Classifier, Business Applications andInterpretation of SVM Models	8
5	**Unit 5: Unsupervised Learning for Business Insights** Principal Component Analysis (PCA),, Principal Components and Dimensionality Reduction, Interpreting PCA for Business Data,, Clustering Techniques: K-Means, Hierarchical Clustering, Choosing the Right Number of Clusters,, Use Cases in CustomerSegmentation, Market Basket Analysis	8
Total Hours		**45**

Textbook :

1 Kumar, U. D. (2020). Business analytics: The science of data-driven decision making(2nd ed.). Wiley India., Kumar, U. D. (2020)., Wiley India., 2020

References:

1 Machine learning for beginners: A non-mathematical guide., Machine learning for beginners: A non-mathematical guide., Misra, D., BPB Publications., 2021

2 Introduction to machine learning., Introduction to machine learning., Ravindran, B. (2019)., McGraw-Hill, 2019

3 Data mining and business intelligence., Data mining and business intelligence., Mitra, S., Oxford University Press., 2019

Suggested Theory Distribution:

The suggested theory distribution as per Bloom's taxonomy is as follows. This distribution serves as guidelines for teachers and students to achieve effective teaching-learning process

Distribution of Theory for course delivery and evaluation

Remember / Knowledge	Understand	Apply	Analyze	Evaluate	Higher order Thinking / Creative
0.00	0.00	35.00	35.00	30.00	0.00

Instructional Method:

1 Class room Teaching

2 case study

Supplementary Resources:

1 https://www.britannica.com/science/statistics/Random-variables-and-probability-distributions

2 https://machinelearningmastery.com/statistical-hypothesis-tests/

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Supplementary Resources:

3 https://www.questionpro.com/features/correlation-analysis.html

4 https://www.sciencedirect.com/topics/medicine-and-dentistry/regression-analysis

5 https://www.aptech.com/blog/introduction-to-the-fundamentals-of-time-series-data-and-analysis

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